# Etienne Meunier

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## Research and Professional Experience

**Aug. 2026 – Sept. 2026** — **Visiting Researcher, Niels Bohr Institute - Copenhagen**

*Development of hybrid ocean models*

Supervisor: Markus Jochum

- Gradient-based optimization in the coupled model VerCOR
- Spin-up approach for Veros with physically constrained neural operators

**Jan. 2025 – Today** — **Postdoc Inria - Paris**

*Machine learning for dynamical systems and differential equations*

Supervisor: Julien Salomon – julien.salomon@inria.fr

- Hybrid models and Neural Differential Equations
- Constrained probabilistic models for physical states generation
- Co-supervision of PhD students M. Janvier and B. Gorce

**Jan. 2024 – Dec. 2024** — **Postdoc CNRS - LOCEAN, Sorbonne Université - Paris**

*Deep learning parametrizations for general model of ocean circulation (OGCMs)*

Supervisor: Julie Deshayes – julie.deshayes@locean.ipsl.fr

- Integration of deep learning closures into large-scale ocean models
- Investigation of optimisation methods for parameter calibration of OGCMs
- International collaborations through the M2Lines initiative

**Sept. 2020 – Nov. 2023** — **PhD in signal, image processing and computer vision - Inria Rennes**

*Unsupervised learning for motion segmentation and motion saliency in videos*

Supervisor: Patrick Bouthemy – patrick.bouthemy@inria.fr

- Mathematical formulation of losses for unsupervised learning
- Development and diffusion of several research publications and code implementations
- Supervision of two master-level interns of 6 months on:
  - → *Efficient computation of optical flow fields on large-scale satellite images*
  - → *Unsupervised segmentation of small moving objects in image sequences*

**Sept. 2019 – Aug. 2020** — **Visiting Scholar, University of British Columbia, Mostafavi Lab - Vancouver**

*Developing computational and statistical approaches for interpreting genomics data*

Supervisor: Sara Mostafavi – saramos@cs.washington.edu

- Developed evolutionary approaches to generate sequences conditioned by the activation of a convolutional neural network

**May 2018 – Aug. 2018** — **Researcher Intern, centre de ressources et d'appui sur le risque requin - La Réunion**

*The CRA is an organization created by the French Government to lead research on innovative solutions to reduce the risk of shark attacks on La Reunion Island (Indian Ocean)*

Supervisor: Eric Chateauminois

- During Internship
  - → Reduced by 10x the amount of videos to watch for biologists by developing a motion detection algorithm that works on underwater videos using several tracking techniques
- As a consultant (Sept. 2018 - Aug. 2020)
  - → Leading the development of shark detection algorithm using a deep learning approach
  - → Supervised the conception of a scientific paper on our domain adaptation strategy for shark detection in the wild

## Teaching

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| **Sept. 2024**<br>**Dec. 2024** | **Machine Learning for Climate and Energy - Ecole Polytechnique, Palaiseau**<br>*Master Level, Instructor* |
| **Sept. 2022**<br>**Dec. 2022** | **Machine learning - EFREI, Paris**<br>*Master Level, Instructor* |
| **Sept. 2021**<br>**Sept. 2022** | **Research methods in computer vision - EFREI, Paris**<br>*Master Level, lecturer for 40+ students* |

## Awards & Distinctions

**Highlight Paper (Top 10%)**, CVPR 2023 — *Unsupervised Space-Time Network for Temporally-Consistent Segmentation of Multiple Motions*

**Best Poster Presentation Award**, Cross VESRI Convening 2024, University of Cambridge — *Diffusion Models for Oceanic States Generation*

## Grants & Funding

**PPR Ocean & Climate initiative** — *Differential Equations in Latent Space for Temporal Acceleration*
174k€, awarded Feb. 2025. Co-PI: J. Salomon (Inria).

**APP2026 – GdR IASIS** — *DIVIN (DIVergence Identiquement Nulle)*
10k€, awarded Feb. 2026. Co-PI: S. Herbreteau (Inria).

## Open Source Software

**DINO-Fusion**
*Diffusion model for generating physically consistent oceanic states*

**Differentiable Veros**
*Contribution to Veros (JAX ocean model) enabling full differentiability for online learning*

**LT-Segmentation**
*Unsupervised motion segmentation: smooth long-term model over a video*

**ST-Space-Time-Flow-Segmentation**
*Unsupervised space-time network for temporally-consistent segmentation of multiple motions*

**EM-Flow-Segmentation**
*EM-driven unsupervised learning for motion segmentation*

**Shape-Checker**
*Minimal package extending einops and providing guardrails for tensor manipulations*

## Education

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|---|---|
| **Jan. 2019**<br>**Sept. 2019** | **Master of Science - Boston University, USA**<br>*Concentration in Computer Information Systems - GPA: 3.96/4*<br>• Research project comparing generative networks and evolutionary approaches |
| **Sept. 2014**<br>**Dec. 2018** | **Diplôme d'ingénieur - ECE Paris, France**<br>*Concentration in Computer Science - Top 5% overall ranking*<br>• Mathematics, Programming, Electronics, Robotics, Data science |

**Reviews for:** IEEE TIP, CVPR, ISBI, ICML, NeurIPS, ICLR, TMLR

# Publications

## Under Review

**ICLR** submitted
**Invertible Continuous Latent Dynamic for Long-Term Data Assimilation in Complex Physical Systems**
R. Grozavescu*, E. Meunier*, P. Zhang & M. Girolami (*equal contribution) [arXiv]

**ICLR** submitted
**Compact Representations of Divergence-Free Fields for Self-Supervised Image Denoising**
S. Herbreteau & E. Meunier [paper]

**JAMES** submitted
**Leveraging Differentiable Programming for Ocean Model Design and Calibration**
S. Ouala, H. Frezat, E. Meunier, R. Fablet & J. Le Sommer [preprint]

**GMD** submitted
**VerCOR v.0.4.6: Quick and Easy Coupling and Prototyping of Earth System Model Components in Python and JAX**
R. Nuterman, I. Rockström, V. Veluvali, E. Meunier, A. L. Stoustrup & M. Jochum

## 2026

**NeurIPS** 2026
**Unrolled Gradients in Disguise: Bridging Interpolation-Based and Jacobian Regularization for Stable Neural Dynamics**
M. Janvier & E. Meunier [paper]

**ICLR** 2026
**Divergence-Free Neural Networks with Application to Image Denoising**
S. Herbreteau & E. Meunier [paper] [code]

**TPAMI** 2026
**Segmenting the Motion Components of a Video: A Long-Term Unsupervised Model**
E. Meunier & P. Bouthemy [paper] [code]

## 2025

**EurIPS-W** 2025
**Towards Fully Differentiable Neural Ocean Model with Veros**
E. Meunier, S. Ouala, H. Frezat, J. Le Sommer & R. Fablet [arXiv] [code]

**NeurIPS-W** 2025
**Physically Consistent Sampling for Ocean Model Initialization**
B. Gorce, L. Ollier, D. Kamm & E. Meunier [paper] [code]

**ICLR-W** 2025
**Learning to Generate Physical Ocean States: Towards Hybrid Climate Modeling**
E. Meunier, D. Kamm, G. Gachon, R. Lguensat & J. Deshayes [arXiv] [code]

## Prior 2025

**CVPR-W** 2024
**Efficient Local Correlation Volume for Unsupervised Optical Flow Estimation on Small Moving Objects in Large Images**
S. Khairi, E. Meunier, R. Fraisse & P. Bouthemy [paper]

**IMWUT** 2024
**ADA-SHARK: A Shark Detection Framework Employing Underwater Cameras and Domain Adversarial Neural Nets**
M. Martin, E. Meunier, P. Moreau, J. Gadenne, J. Dautel, E. Pinsky, ... & R. Rawassizadeh

**CVPR** 2023 Highlight
**Unsupervised Space-Time Network for Temporally-Consistent Segmentation of Multiple Motions**
E. Meunier & P. Bouthemy [paper] [code]

**TPAMI** 2023
**EM-driven Unsupervised Learning for Motion Segmentation**
E. Meunier, A. Badoual & P. Bouthemy [paper] [code]

**BMVC** 2021
**Unsupervised Computation of Salient Motion Maps from the Interpretation of a Frame-Based Classification Network**
E. Meunier & P. Bouthemy [paper]

**MLCB** 2020
**Convolutional Additive Models: a Fully Interpretable Approach to Deep Learning in Genomics**
G. Novakovsky, M. Saraswat, E. Meunier, O. Fornes, S. Mostafavi & W. Wasserman

**MLCB** 2019
**Interpreting Deep Learning Models in Genomics using Genetic Algorithm**
E. Meunier, G. Novakovsky & S. Mostafavi